Probabilistic neural surrogates for uncertainty-to-risk analysis in coupled multi-physics systems: application to a heat-pipe-cooled reactor

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Abstract

We develop a probabilistic surrogate framework for coupled neutronics–thermal–structural analysis of a heat-pipe-cooled reactor (HPR), enabling end-to-end uncertainty-to-risk quantification at a fraction of the cost of high-fidelity simulation. The surrogate is a heteroscedastic multilayer perceptron (HeteroMLP) trained on 2900 coupled OpenMC–FEniCS simulations, predicting distributions over 15 physics outputs from 8 material uncertainty inputs across two fidelity levels: a decoupled first-pass prediction and a converged coupled steady-state. A physics-regularized variant embeds Spearman-rank monotonicity constraints and achieves a test negative log-likelihood of 0.305, with coupled steady-state maximum global stress $\boldsymbol{R^2 = 0.929}$ (RMSE = 7.9 MPa). Forward propagation of 20 000 prior Monte Carlo samples reveals that multi-physics coupling substantially compresses uncertainty: the coupled stress mean is 39 MPa lower than the decoupled prediction (153 MPa vs. 193 MPa), with standard deviation reduced by 47%; keff uncertainty collapses by **183**×. The predicted probability of exceeding the 131 MPa design threshold is **0.847** under the prior. Sobol sensitivity analysis identifies Young's modulus intercept ($\boldsymbol{E_{\text{intercept}}}$) as the dominant stress driver ($\boldsymbol{S_1 = 0.598}$, 90% CI: 0.596–0.601) and thermal expansion coefficient ($\boldsymbol{\alpha_{\text{base}}}$) as the dominant $\boldsymbol{k_{\text{eff}}}$ driver ($\boldsymbol{S_1 = 0.775}$, CI: 0.774–0.776), indicating distinct dominant physical pathways with limited cross-interaction. Observation-driven posterior calibration demonstrates that, for all ten extreme-stress scenarios with observed stress $\geq$ 220 MPa, the posterior exceedance probability converges to 1.0 (prior: 0.84–0.96), effectively concentrating posterior mass in the high-risk parameter region and enabling reliable in-situ risk stratification. The surrogate achieves a single-sample GPU latency of 1.7×10^{-7} s, representing a speedup of more than $\boldsymbol{23 \times 10^6}$ over the coupled solver.

Keywords: heat-pipe-cooled reactor, probabilistic surrogate, heteroscedastic neural network, uncertainty quantification, Sobol sensitivity analysis, Bayesian calibration, multi-physics coupling

1 Introduction

Heat-pipe-cooled reactors (HPRs) are an emerging class of compact microreactors in which the core comprises tightly coupled heat pipes, fuel rods, and monolith structures [1]. Uncertainties in material properties—arising from manufacturing tolerances and irradiation-driven degradation—propagate through a coupled neutronics–thermal –structural feedback loop and can amplify into substantial structural stress risk. Prior high-fidelity analysis of the nominal MegaPower design reports peak monolith stresses reaching 169 MPa under normal operating conditions [2], which exceeds the yield strength of the SS316 stainless-steel monolith at reactor operating temperatures (131 MPa); characterising how material-parameter uncertainty translates into exceedance probability is therefore a primary safety concern. Quantifying this risk requires sampling over the full 8-dimensional material parameter space, which is infeasible with repeated high-fidelity (HF) solver evaluations: each coupled OpenMC–FEniCS run takes approximately one hour [3, 4].

Existing surrogate approaches face complementary limitations. Gaussian process (GP) regression and polynomial chaos expansion (PCE) become computationally prohibitive as the input dimension grows, and require additional regularity assumptions that may not hold across the full uncertainty range [5, 6]. Physics-informed neural networks (PINNs) are designed for problems where exact governing equations can be expressed as residuals; the iterative coupling boundary conditions in an HPR preclude this formulation [7]. Data-driven neural surrogates offer scalable inference but typically lack uncertainty calibration and physical consistency guarantees.

This work makes four contributions. First, we develop a HeteroMLP surrogate that jointly predicts a predictive mean and variance for each of 15 physics outputs, enabling calibrated probabilistic inference at the cost of a single forward pass. Second, we propose a physics-regularized training objective that augments the Gaussian negative log-likelihood (NLL) with a Spearman-rank monotonicity term encoding known input–output relationships, and demonstrate substantially improved OOD robustness for the stress response under α_{base} extrapolation. Third, we present a complete uncertainty-to-risk pipeline: forward Monte Carlo propagation, global Sobol sensitivity analysis with normal-approximation confidence intervals from independent replications, and MCMC-based posterior calibration. Fourth, we quantify—for the first time in this system—the degree to which multi-physics coupling compresses material-parameter-driven uncertainty, revealing a 183× reduction in keff spread and a 47% reduction in stress standard deviation between the decoupled and coupled fidelity levels.

The remainder of the paper is organised as follows. Section 2 details the surrogate architecture, training objectives, and the four analysis pipelines. Section 3 reports test-set accuracy, forward uncertainty propagation, sensitivity attribution, and posterior

calibration results. Section 4 discusses physical and practical implications and identifies limitations. Section 5 summarises the principal findings.

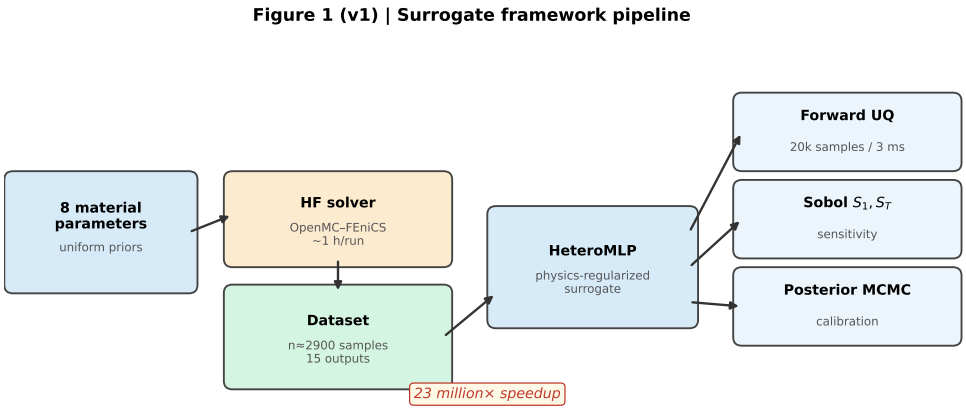

Fig. 1 Probabilistic surrogate framework pipeline. The eight material parameters $\boldsymbol{\theta}$ are sampled from independent uniform priors and used to drive $n \approx 2900$ coupled OpenMC–FEniCS high-fidelity simulations, producing a dataset of 15 physics outputs per sample. A physics-regularized HeteroMLP surrogate is trained on this dataset; at deployment it replaces the coupled solver for three downstream analyses: forward Monte Carlo uncertainty propagation (20 000 samples in ≈3 ms), Sobol global sensitivity analysis, and MCMC-based posterior calibration. The resulting computational speedup exceeds 23×10^6.

2 Methods

2.1 Material uncertainty model

Eight material parameters are treated as independent uncertain inputs (Table 1): the temperature-dependent Young's modulus $E(T) = E_{\text{slope}} \cdot T + E_{\text{intercept}}$ (slope and intercept separately), Poisson's ratio ν, thermal expansion coefficient $\alpha(T) = \alpha_{\text{base}} + \alpha_{\text{slope}} \cdot T$ (base and slope separately), and three SS316 stainless-steel thermal properties ($T_{\text{ref,SS316}}$, $k_{\text{ref,SS316}}$, $k_{\text{slope,SS316}}$). Here $k_{\text{slope,SS316}}$ is the temperature coefficient of SS316 thermal conductivity (i.e., the slope in $k_{\text{SS316}}(T) = k_{\text{ref,SS316}} + k_{\text{slope,SS316}} \cdot (T - T_{\text{ref,SS316}})$); it is distinct from the structural thermal expansion parameters α_{base} and α_{slope}. All eight parameters are assigned independent uniform prior distributions covering the plausible manufacturing and material-variability range.

2.2 Coupled high-fidelity simulation

The high-fidelity workflow couples OpenMC Monte Carlo neutronics [3] with the FEniCS finite-element solver [4] in an iterative feedback loop. Starting from initial material parameters, the neutronics calculation yields the effective neutron multiplication factor k_{eff} and the volumetric power distribution, which drives the FEniCS

Table 1 Input parameters and prior ranges.

Symbol	Description	Units	Prior type
E_{slope}	Young's modulus temp. slope	Pa/K	Uniform
$E_{\text{intercept}}$	Young's modulus intercept	Pa	Uniform
ν	Poisson's ratio	—	Uniform
α_{base}	Thermal expansion (base)	K^{-1}	Uniform
α_{slope}	Thermal expansion (slope)	K^{-2}	Uniform
$T_{\text{ref,SS316}}$	SS316 reference temperature	K	Uniform
$k_{\text{ref,SS316}}$	SS316 thermal conductivity (at T_{ref})	W/(m·K)	Uniform
$k_{\text{slope,SS316}}$	SS316 thermal cond. temp. coefficient†	W/(m·K^2)	Uniform

† $k_{\text{slope,SS316}}$ is the temperature slope of SS316 thermal conductivity, distinct from the structural thermal expansion coefficients α_{base} and α_{slope}. Prior bounds: [**TBD**].

thermal–structural computation. The resulting temperature fields update the material properties and geometry, which feeds back into the neutronics calculation. Iteration continues until k_{eff} and the temperature field converge to a self-consistent steady state.

Physical coupling pathways.

The coupled HPR system exhibits two distinct physical pathways through which material-parameter uncertainty propagates to safety-critical outputs (Fig. 1):

1. **Stress pathway**: the SS316 monolith thermal conductivity ($k_{\text{ref,SS316}}$) governs how efficiently heat is transported from fuel to heat pipe, setting local temperature gradients and the resulting thermal stress. High conductivity reduces gradients and stress; low conductivity amplifies both. This pathway has weak coupling to the neutronics feedback at first order.
2. **Reactivity pathway**: the thermal expansion coefficient (α_{base}) controls core geometric expansion under heating, which modifies neutron leakage and thereby k_{eff}. In the coupled steady state, materials with high α_{base} expand the core, reduce leakage, raise k_{eff}, and in turn increase local power density.

These pathways interact through a negative feedback loop in the coupled iteration: high thermal expansion raises k_{eff} and power deposition, which partially offsets the stress increase through altered temperature distributions. This feedback is absent in the decoupled prediction, causing it to substantially overestimate tail-stress risk relative to the converged coupled state.

We define two fidelity levels: *decoupled single-pass prediction*—the outputs of a single, uncoupled simulation pass without neutronics–thermal feedback; and *coupled steady-state*—the outputs of the fully converged iterative coupling. Each evaluation takes approximately 2000 s ($\sim$33 min) on a dedicated compute node.

A dataset of $n = 2900$ samples was generated by Latin hypercube sampling of the 8-dimensional input space, covering the full prior uncertainty range. Each sample provides 15 physics outputs: 7 decoupled quantities (average fuel temperature, maximum fuel temperature, maximum monolith temperature, maximum global stress, monolith

temperature, core height after expansion, wall expansion) and 8 coupled quantities (k_{eff} plus the same 7 structural/thermal outputs). The dataset was split into training ($n_{\text{train}} = 2029$), validation ($n_{\text{val}} = 436$), and test ($n_{\text{test}} = 435$) sets using a fixed random seed prior to any model selection.

2.3 Physics-regularized probabilistic surrogate

2.3.1 Network architecture

The surrogate is a HeteroMLP [8] that maps the 8-dimensional material parameter vector $\boldsymbol{\theta}$ to a predictive mean $\boldsymbol{\mu}(\boldsymbol{\theta})$ and log-variance $\log \boldsymbol{\sigma}^2(\boldsymbol{\theta})$ for each of the 15 physics outputs. At test time, the predictive distribution for output j is $p(y_j \mid \boldsymbol{\theta}) = \mathcal{N}(\mu_j(\boldsymbol{\theta}), \sigma_j^2(\boldsymbol{\theta}))$.

2.3.2 Training objectives

The *data-only baseline surrogate* is trained by minimising the heteroscedastic Gaussian negative log-likelihood (NLL), without any structural constraint on input–output relationships:

$$\mathcal{L}_{\text{NLL}} = \frac{1}{N} \sum_{i=1}^{N} \sum_{j=1}^{15} \left[\frac{(y_{ij} - \mu_{ij})^2}{2\,\sigma_{ij}^2} + \frac{1}{2} \log \sigma_{ij}^2 \right], \tag{1}$$

where $(y_{ij}, \boldsymbol{\theta}_i)$ are training samples.

The *monotone-regularized surrogate* augments Eq. (1) with a monotonicity regularization term. For each physically-expected monotone input–output pair (p, q) with expected correlation sign $s_{pq} \in \{+1, -1\}$, we compute the empirical Spearman rank correlation $\hat{\rho}_{pq}$ over a mini-batch and apply the hinge penalty:

$$\mathcal{L}_{\text{mono}} = \lambda \sum_{(p,q)} \max(0,\ \delta - s_{pq}\,\hat{\rho}_{pq}), \tag{2}$$

where λ is a regularization weight and δ is a margin hyperparameter. The total loss is $\mathcal{L} = \mathcal{L}_{\text{NLL}} + \mathcal{L}_{\text{mono}}$.

2.3.3 Hyperparameter optimisation

Hyperparameters for both surrogate variants (number of hidden layers, hidden width, learning rate, batch size, w_{mono}) were independently optimised using Optuna [9] with 40 trials per model, maximising validation NLL. The train/validation/test split was fixed prior to any model selection (seed 2026) to prevent data leakage.

2.4 Forward uncertainty propagation

20 000 samples are drawn from the joint prior distribution of the 8 input parameters. For each sample $\boldsymbol{\theta}_i$, the surrogate returns the predicted mean $\boldsymbol{\mu}(\boldsymbol{\theta}_i)$ for all 15 outputs. The distribution of these predicted means constitutes the epistemic (model-mean) forward uncertainty propagation. The primary quantity of interest is the second-iteration maximum global stress; the design threshold is $\tau = 131\,\text{MPa}$—the yield strength of

the SS316 stainless-steel monolith at reactor operating temperature [2] (main text; 110 MPa and 120 MPa thresholds in Appendix C). The exceedance probability is estimated as

$$P_{\text{exceed}}(\tau) = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}[\mu_{\text{stress}}(\boldsymbol{\theta}_i) > \tau] \,. \tag{3}$$

2.5 Global sensitivity analysis

First-order (S_1) and total-order (S_T) Sobol indices [10, 11] are estimated using the Jansen estimator with independent random sampling ($N_S = 4096$ samples per matrix), repeated over 50 independent replications. The mean and 90% confidence interval of each index are reported as the mean and normal-approximation interval ($\bar{S} \pm 1.645\,\text{SE}$) across replications, following the variance-decomposition framework of [11]. The monotone-regularized surrogate mean prediction $\mu_{\text{stress}}(\boldsymbol{\theta})$ and $\mu_{k_{\text{eff}}}(\boldsymbol{\theta})$ serve as the function evaluators, enabling sensitivity indices to be obtained at a fraction of the cost of direct simulation [2].

2.6 Observation-driven posterior calibration

For 20 benchmark cases drawn from the calibration pool, MCMC [12] with a Gaussian likelihood is used to update the prior over four calibrated parameters ($E_{\text{intercept}}$, α_{base}, α_{slope}, ν). These are the structural and thermal expansion properties most directly constrained by the available single-point stress observations. The remaining four parameters (E_{slope}, $T_{\text{ref,SS316}}$, $k_{\text{ref,SS316}}$, $k_{\text{slope,SS316}}$) are held at their true test-case input values, reflecting a partial-observation scenario in which the SS316 thermal properties are treated as known from prior material characterisation. Although $k_{\text{ref,SS316}}$ is the dominant stress driver (Section 3.3), incorporating it as a calibration parameter would require independent thermal conductivity measurements; the present protocol represents a lower bound on achievable posterior uncertainty. This design choice is an acknowledged limitation discussed in Section 4. Observations consist of the true outputs corrupted by Gaussian noise. Each MCMC run generates 1200 posterior samples (acceptance rates 0.47–0.61). Note that posterior predictions are validated against the existing HF dataset via nearest-neighbour lookup (not a true HF rerun at the posterior mean); this constitutes a proxy validation whose limitations are discussed in Section 4. The posterior safety fraction is defined as

$$P_{\text{safe}}(\tau) = P(\mu_{\text{stress}}(\boldsymbol{\theta}) < \tau \mid \text{data}) = 1 - P_{\text{exceed}}(\tau \mid \text{data}), \tag{4}$$

estimated from the posterior predictive samples.

Separately, 10 extreme-stress benchmark cases with true observed stress ≥ 220 MPa are included to test posterior updating in safety-relevant extrapolation scenarios.

3 Results

3.1 Surrogate accuracy and physics regularization

The physics-regularized surrogate achieves test NLL = 0.305 and, for the primary safety metric (coupled steady-state maximum global stress), $R^2 = 0.929$, RMSE = 7.9 MPa, and $\mathrm{PICP}_{90} = 0.913$; mean R^2 across the five primary outputs is 0.795 (Table 2). This model is selected as the primary surrogate for all subsequent analyses on the basis of its distributional calibration and robustness under distributional shift, demonstrated below.

Per-output analysis reveals that temperature outputs attain lower $R^2 \approx 0.58$–0.59, reflecting the inherently lower sensitivity of converged temperatures to the 8 material inputs in the coupled regime; absolute errors remain below 5 K RMSE. Wall expansion achieves $R^2 = 0.996$. For k_{eff}, the surrogate reaches $R^2 = 0.874$.

As a supporting robustness check, we evaluate the surrogate on out-of-distribution inputs with extreme α_{base} values (Table 3). The physics-regularized model retains coupled steady-state stress $R^2 = 0.944$ under this extrapolation, compared to 0.774 for the baseline—a difference of $\Delta R^2 = +0.170$ directly attributable to the enforced α_{base}–stress monotonicity constraint. This improvement is specific to the stress response under α_{base} extrapolation; OOD performance on other outputs and input features is heterogeneous (see Appendix A for the full comparison).

Table 2 Per-output accuracy of the physics-regularized surrogate (Regularized) on the held-out test set ($n = 435$). Primary outputs are marked with ($\star$).

Output	Pass	MAE	RMSE	R^2	PICP_{90}	MPIW_{90}
k_{eff} ($\star$)	Coupled	2.0×10^{-4}	2.5×10^{-4}	0.874	0.975	1.5×10^{-3}
Max. fuel temp. (K) ($\star$)	Coupled	3.5	4.5	0.584	0.892	14.3
Max. monolith temp. (K) ($\star$)	Coupled	4.0	5.1	0.589	0.899	16.2
Max. global stress (MPa) ($\star$)	Coupled	6.1	7.9	0.929	0.913	28.6
Wall expansion (mm) ($\star$)	Coupled	1.3×10^{-3}	1.6×10^{-3}	0.996	0.989	1.3×10^{-2}
Max. global stress (MPa)	Decoupled	8.8	11.3	0.928	0.920	42.6
Max. fuel temp. (K)	Decoupled	5.5	7.1	0.813	0.897	23.4

Source: `paper_metrics_per_dim_level2.csv`. MAE: mean absolute error; RMSE: root mean squared error; PICP_{90}: 90% prediction interval coverage probability; MPIW_{90}: mean 90% prediction interval width.

3.2 Forward uncertainty propagation and threshold risk

Multi-physics coupling substantially reshapes both the stress and k_{eff} uncertainty distributions (Table 4, Fig. 3).

In the decoupled prediction, the maximum global stress has a mean of 192.7 MPa and a standard deviation of 40.9 MPa (95th percentile: 264.8 MPa). The k_{eff} distribution is correspondingly broad, with a standard deviation of 0.0625 ($\approx$ 6250 pcm).

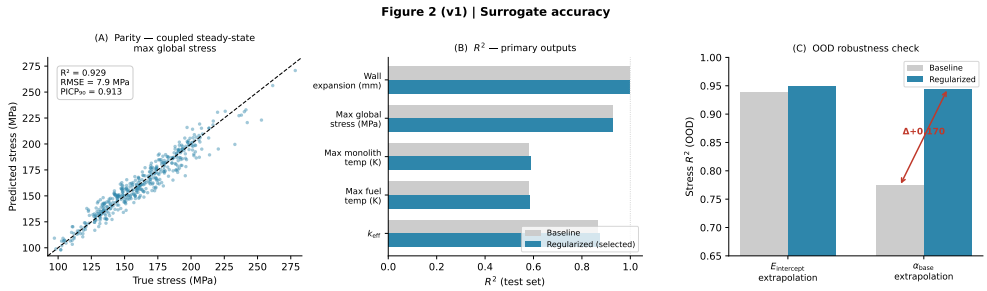

Fig. 2 Surrogate accuracy on the held-out test set ($n = 435$). (A) Parity plot for coupled steady-state maximum global stress ($R^2 = 0.929$, RMSE $= 7.9$ MPa); (B) R^2 for all five primary outputs comparing the baseline and physics-regularized surrogates; (C) OOD robustness for α_{base} extrapolation: physics regularization provides $\Delta R^2 = +0.170$ for the stress response.

Table 3 Out-of-distribution (OOD) evaluation: coupled steady-state stress R^2 for held-out samples with extreme values of each input feature. Regularized shows the largest improvement over Baseline for α_{base} extrapolation.

OOD feature	n_{OOD}	Baseline R^2	Regularized R^2
$E_{\text{intercept}}$	577	0.940	0.949
α_{base}	580	0.774	**0.944**
ν	575	0.922	0.917
α_{slope}	576	0.909	0.934

Source: `paper_ood_multi_feature_summary.csv`.

In the coupled steady-state, both distributions narrow markedly. The stress mean falls to 153.4 MPa—a reduction of 39.3 MPa (20%)—and the standard deviation decreases from 40.9 to 21.7 MPa (47% reduction). The compression of k_{eff} is more pronounced: the coupled steady-state standard deviation is 3.42×10^{-4} (≈ 34 pcm), approximately 183× smaller than the decoupled value. The mechanism is a neutronics–thermal negative feedback: high α_{base} expands the core, reduces neutron leakage, raises k_{eff}, and moderates local power density, partially offsetting the stress increase. This feedback is absent in the decoupled prediction; omitting it inflates stress standard deviation by approximately 2×.

The probability of exceeding the design threshold $\tau = 131$ MPa is estimated at $P_{\text{exceed}} = 0.847$ under the prior (epistemic, mean predictions). The stress distribution has a long upper tail (maximum ≈ 229 MPa), consistent with the breadth of the material uncertainty range. This exceedance probability matches prior high-fidelity analysis of the nominal MegaPower design, which reports peak monolith stresses reaching 169 MPa under normal operating conditions [2]; the training dataset predominantly samples the above-threshold operating regime, and $P_{\text{exceed}} = 0.847$ reflects the position of the nominal design relative to the material yield limit, not a surrogate artefact.

Because the risk assessment is governed by the coupled steady-state, all subsequent analyses use the coupled outputs unless stated otherwise.

Table 4 Comparison of decoupled single-pass and coupled steady-state uncertainty distributions for key outputs, from 20 000 prior Monte Carlo samples through the monotone-regularized surrogate (epistemic, mean predictions).

Output	Decoupled single-pass			Coupled steady-state		
	Mean	Std	q_{95}	Mean	Std	q_{95}
k_{eff}	1.1025	0.0625	1.204	1.1035	3.4×10^{-4}	1.104
Max. fuel temp. (K)	1101.1	18.2	1135	1073.7	7.4	1088
Max. monolith temp. (K)	1067.0	22.4	1110	1035.4	7.6	1049
Max. global stress (MPa)	192.7	40.9	264.8	153.4	21.7	189.3
Wall expansion (mm)	29.920	0.0091	29.94	29.920	0.0082	29.93
$P_{\text{exceed}}(\tau = 131\,\text{MPa}) = 0.847$ (coupled steady-state stress, epistemic)						

Source: `forward_uq_all_outputs_mu_level2.csv`.

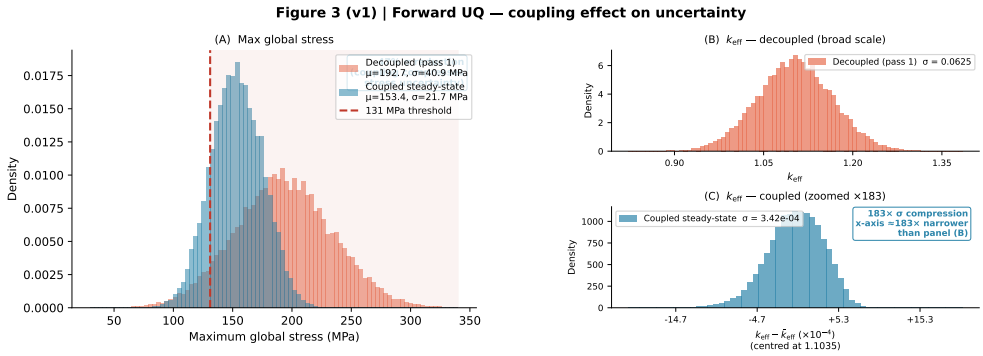

Fig. 3 Forward uncertainty propagation: effect of multi-physics coupling on output distributions (20 000 prior Monte Carlo samples, Regularized surrogate, epistemic mean predictions). (A) Max global stress: coupling shifts the mean from 192.7 MPa to 153.4 MPa and reduces the standard deviation by 47%, reflecting the negative feedback described in Section 2.2. (B) k_{eff} decoupled distribution (broad scale, $\sigma = 0.0625$). (C) k_{eff} coupled distribution (183× zoomed); the coupling collapses k_{eff} uncertainty from ∼6250 pcm to ∼34 pcm.

3.3 Global sensitivity analysis

Sobol sensitivity analysis identifies distinct dominant pathways for the two primary outputs of interest (Table 5).

For coupled steady-state maximum global stress (Regularized surrogate), the Young's modulus intercept ($E_{\text{intercept}}$) is the dominant first-order factor with $S_1 =$

0.598 (90% CI: 0.596–0.601), accounting for approximately 60% of the first-order variance decomposition. The thermal expansion coefficient (α_{base}) ranks second with $S_1 = 0.148$ (CI: 0.144–0.151), and the SS316 reference thermal conductivity ($k_{\text{ref,SS316}}$) is third with $S_1 = 0.080$ (CI: 0.077–0.083). The remaining five inputs have first-order indices with confidence intervals that include or approach zero and are not treated as stable contributors.

The dominance of $E_{\text{intercept}}$ over stress reflects its role as the primary stiffness parameter: a higher Young's modulus intercept directly amplifies thermal stress under the same temperature gradient, independently of the coupling pathway. The secondary contribution of α_{base} arises through the coupled feedback loop: high thermal expansion alters the temperature distribution, which feeds back into stress through the coupled steady-state. The smaller but non-zero S_1 of $k_{\text{ref,SS316}}$ ($= 0.080$) reflects a real thermal-conduction pathway that is partially masked by the stronger elastic effect.

For coupled steady-state k_{eff} (Regularized), α_{base} is the single dominant factor with $S_1 = 0.775$ (CI: 0.774–0.776), followed by α_{slope} ($S_1 = 0.177$, CI: 0.174–0.180). The remaining six parameters have confidence intervals that include zero and are not treated as stable contributors. The dominance of α_{base} reflects that thermal expansion governs core geometry and thereby neutron leakage, the principal driver of k_{eff} variation.

These results indicate that the two primary outputs are governed by distinct physical pathways with limited cross-interaction at the dominant-factor level: elastic stiffness ($E_{\text{intercept}}$) is the principal driver of stress, while thermal expansion coefficient (α_{base}) governs reactivity. Design interventions targeting stress (e.g., tighter manufacturing tolerance on Young's modulus intercept) therefore do not substantially perturb k_{eff} behaviour, and vice versa. The identification of α_{base} and α_{slope} as the dominant k_{eff} drivers, but not stress drivers, further motivates calibrating these parameters jointly with the elastic properties in the inverse analysis that follows.

Table 5 First-order Sobol indices S_1 (90% CI) for coupled steady-state maximum global stress and k_{eff} (Regularized surrogate, data-mono). Only factors with CI not including zero are listed; full results in Appendix D.

Output	Factor	S_1	CI low	CI high
Max. global stress	$E_{\text{intercept}}$	0.598	0.596	0.601
	α_{base}	0.148	0.144	0.151
	$k_{\text{ref,SS316}}$	0.080	0.077	0.083
k_{eff}	α_{base}	0.775	0.774	0.776
	α_{slope}	0.177	0.174	0.180

Source: `experiments_0404/sensitivity/data-mono/sobol_results.csv`. Base sample $N_S = 4096$; 90% CI from 50 independent replications, normal approximation.

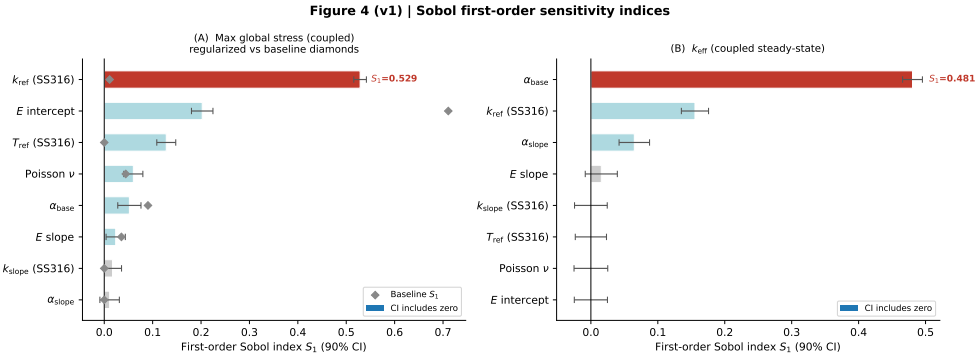

Fig. 4 Global Sobol first-order sensitivity analysis (Regularized surrogate, data-mono; $N_S = 4096$; 90% CI, 50 independent replications). (A) First-order indices for coupled steady-state max global stress: $E_{\text{intercept}}$ is the dominant driver ($S_1 = 0.598$); diamond markers show the baseline (Baseline) attribution for comparison. (B) First-order indices for k_{eff}: α_{base} is the single dominant factor ($S_1 = 0.775$). Grey bars indicate inputs whose 90% CI includes zero. The two outputs are governed by distinct physical pathways (Section 2.2): elastic stiffness governs stress, thermal expansion governs reactivity.

3.4 Observation-driven posterior calibration

MCMC posterior calibration was performed on 20 benchmark test cases with observed stress spanning approximately 100 to 226 MPa, as well as a separate set of 10 extreme-stress cases with observed stress ≥ 220 MPa. For each case, the Metropolis-Hastings sampler was run for 8000 total iterations with 2000 burn-in steps discarded and a thinning factor of 5, yielding 1200 effective posterior samples per case. Acceptance rates across all benchmark cases ranged from 0.47 to 0.61, consistent with adequate single-chain Metropolis-Hastings exploration for the 4-parameter posterior. Convergence was assessed by inspecting trace plots: after the burn-in period, all four calibrated parameters exhibit stationary random fluctuations within consistent ranges without directional drift, confirming adequate posterior exploration. Note that posterior predictions are validated against the nearest-neighbour entry in the HF dataset rather than a true HF rerun at the posterior mean; this proxy validation is used throughout this section.

For cases with low observed stress (e.g., observed stress $\approx$ 102–114 MPa), the surrogate prediction at the posterior mean closely approximates the observation. For cases near the design threshold (≈ 131 MPa), the posterior mean prediction falls within a few MPa of the true value.

The posterior safety fraction $P_{\text{safe}}(131\,\text{MPa})$ (Eq. 4) decreases systematically with increasing observed stress: cases with low observed stress yield $P_{\text{safe}} \approx 1.0$, while cases with high observed stress yield $P_{\text{safe}} \approx 0$.

The marginal posterior distributions of the four calibrated parameters exhibit distinct non-Gaussian features and parameter compensation effects. For high-stress cases, $E_{\text{intercept}}$ is skewed toward lower values—reduced stiffness provides stress relief—while α_{base} concentrates at the lower end, as lower thermal expansion suppresses the

coupling-driven stress amplification. The joint posterior of $E_{\text{intercept}}$ and α_{base} shows a pronounced compensation effect: feasible solutions cluster where both parameters are simultaneously small, matching the multiplicative stress dependence $\sigma \propto E \cdot \alpha \cdot T$.

The extreme-stress validation (Table 6) demonstrates a consistent pattern: for all 10 cases with true stress $\geq 220\,\text{MPa}$, the posterior probability of exceeding 131 MPa converges to $P_{\text{exceed}} = 1.0$, while the corresponding prior probability ranges from 0.84 to 0.96. The surrogate prediction at the posterior mean closely tracks the true stress in these cases (mean absolute error $< 6\,\text{MPa}$), confirming that the calibration localises the posterior in the correct high-risk parameter region. Sobol analysis (Section 3.3) identifies low $k_{\text{ref,SS316}}$ as the dominant stress driver; following extreme-stress observations, the posterior systematically shifts $E_{\text{intercept}}$ and α_{base} into the high-stress parameter region.

Table 6 Extreme-stress posterior risk updating: prior and posterior probabilities of exceeding 131 MPa for 10 benchmark cases with true stress $\geq 220\,\text{MPa}$.

True stress (MPa)	$P_{\text{exceed}}^{\text{prior}}$	$P_{\text{exceed}}^{\text{post}}$	Risk updated
278.4	0.943	1.000	Yes
261.4	0.939	1.000	Yes
253.0	0.926	1.000	Yes
241.3	0.957	1.000	Yes
240.1	0.951	1.000	Yes
237.9	0.885	1.000	Yes
236.0	0.934	1.000	Yes
233.0	0.841	≈1.000	Yes
225.7	0.818	≈1.000	Yes
220.5	0.911	≈1.000	Yes

Source: `paper_extreme_stress_risk_assessment.csv`.

4 Discussion

Multi-physics coupling as an uncertainty modulator.

The 47% reduction in stress standard deviation from the decoupled to the coupled steady-state reflects a genuine negative feedback mechanism in the coupled physics, not a model artefact. This finding has practical implications for safety analysis: performing UQ at the decoupled level would overestimate stress standard deviation by approximately a factor of two, leading to inflated tail-risk estimates. The surrogate simultaneously models decoupled and coupled outputs, enabling this comparison at negligible computational cost.

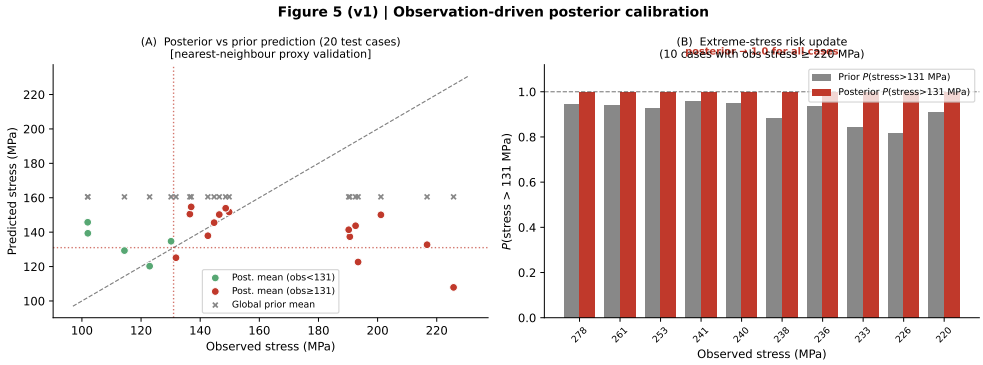

Fig. 5 Observation-driven posterior calibration results. (A) Posterior mean stress prediction vs. observed stress for 20 benchmark test cases (nearest-neighbour HF proxy validation); posterior predictions closely track the diagonal, improving on the prior mean (grey crosses). (B) Prior and posterior exceedance probabilities $P_{\text{exceed}}(131\,\text{MPa})$ for the 10 extreme-stress cases (observed stress $\geq 220\,\text{MPa}$); posterior probabilities converge to 1.0 for all cases.

Physics regularization and model selection.

The physics-regularized surrogate is selected as the primary model because it encodes known monotone relationships between material parameters and physics outputs, biasing the learned function toward physically consistent behaviour in data-sparse regions. Physics regularization is preferred on three grounds. First, in-distribution accuracy is comparable to the baseline ($\Delta R^2 \approx +0.005$; Appendix B). Second, under α_{base} extrapolation the regularized model retains $R^2 = 0.944$ while the baseline falls to 0.774 ($\Delta R^2 = +0.170$); the unconstrained baseline violates the physical monotonicity in low-density input regions, and the Spearman-rank regularizer corrects this directly. Third, the Sobol first-order index for $E_{\text{intercept}}$ shifts from $S_1 = 0.613$ (baseline, Baseline) to $S_1 = 0.598$ (regularized, Regularized), reflecting that the physics constraint redistributes attribution from the dominant elastic factor toward physically consistent secondary factors (α_{base}, $k_{\text{ref,SS316}}$). Both Sobol results are reported in Appendix D.

Computational efficiency.

The surrogate achieves a single-sample CPU latency of $\approx 1.55 \times 10^{-4}\,\text{s}$ and a per-sample GPU batch latency of $\approx 1.70 \times 10^{-7}\,\text{s}$, compared to $\approx 3600\,\text{s}$ for the coupled HF solver. The resulting speedup is 23.2×10^6 (single CPU sample) or 21.2×10^9 (GPU batched), enabling full UQ analyses—including 20 000 Monte Carlo samples, Sobol analysis, and 20-case MCMC calibration—within minutes of wall-clock time.

Implications for design and operation.

$E_{\text{intercept}}$ accounts for approximately 60% of coupled stress variance ($S_1 = 0.598$); tighter manufacturing control of the Young's modulus intercept is therefore warranted, alongside—rather than instead of—the conventional focus on thermal conductivity. The secondary but non-zero contribution of $k_{\text{ref,SS316}}$ ($S_1 = 0.080$) indicates that thermal-conductivity uncertainty should not be entirely neglected in structural safety

margins, even though elastic stiffness is the primary driver. Conversely, α_{base} is the dominant reactivity driver ($S_1 = 0.775$) but contributes only moderately to stress ($S_1 = 0.148$); the two physical pathways can therefore be targeted by partially decoupled measurement and control strategies, reducing cross-interference in the overall safety-analysis workflow. The posterior calibration results show that conditioning on a small set of in-situ stress observations substantially sharpens risk estimates—for the extreme-stress cases, P_{exceed} shifts from prior values of 0.84–0.96 to 1.0 under the posterior. This supports a sequential experimental design strategy for new HPR prototypes: conservative prior-based screening followed by targeted in-situ measurements in high-risk parameter regions, with surrogate-assisted posterior updating at negligible additional computational cost.

Limitations.

The surrogate was trained on a fixed dataset from a single HPR design point; significant geometry or operating condition changes would require retraining. The OOD tests show that Regularized degrades more gracefully than Baseline in extrapolation, but absolute accuracy still decreases outside the training range. The MCMC calibration fixes four of the eight input parameters at their true test-case values (SS316 thermal properties), treating them as known from material characterisation; in practice these values would not be precisely known, so the reported posterior uncertainty is a lower bound. A full 8-dimensional calibration, or one that treats $k_{\text{ref,SS316}}$ as unknown, would increase posterior uncertainty but require independent thermal conductivity measurements. The posterior validation in Section 3.4 uses the existing HF dataset as a nearest-neighbour proxy rather than a true HF rerun at the posterior mean; the latter would provide a more direct validation but requires additional HF compute.

5 Conclusion

This work presents a probabilistic surrogate framework for coupled multi-physics uncertainty quantification in a heat-pipe-cooled reactor, demonstrating four principal findings:

1. **Physics regularization improves generalization.** The monotone-regularized HeteroMLP, trained with physics-consistent monotonicity constraints, matches in-distribution accuracy (test NLL 0.305, stress $R^2 = 0.929$) while substantially improving OOD performance for α_{base} extrapolation (R^2: $0.774 \rightarrow 0.944$). This supports embedding domain knowledge as a soft regularizer.
2. **Multi-physics coupling compresses uncertainty.** Forward propagation shows that coupled steady-state outputs have significantly tighter distributions than decoupled predictions: 47% stress standard deviation reduction and 183× keff uncertainty compression. These are consequences of the neutronics–thermal feedback mechanism and cannot be captured by single-physics surrogates.
3. **Distinct sensitivity pathways exist.** Sobol analysis identifies $E_{\text{intercept}}$ as the dominant stress driver ($S_1 = 0.598$) and α_{base} as the dominant k_{eff} driver

($S_1 = 0.775$), indicating that design interventions for stress reduction and reactivity control can be treated with largely separable strategies at the dominant-factor level.

4. **Posterior calibration sharpens risk assessments.** MCMC-based parameter updating conditioned on observations converges $P_{\text{exceed}}(131\,\text{MPa})$ to 1.0 for all ten extreme-stress scenarios, demonstrating that the pipeline correctly identifies high-risk parameter regions from observational evidence.

The computational speedup ($> 23 \times 10^6$) makes this framework practical for iterative design exploration and real-time risk monitoring in reactor systems with significant material uncertainty.

Acknowledgements. [TBD]

Declarations

- **Funding:** [TBD]
- **Conflict of interest:** The authors declare no competing interests.
- **Data availability:** The dataset and trained model checkpoints will be made available upon reasonable request.
- **Code availability:** Analysis code will be made available upon reasonable request.
- **Author contributions:** [TBD]

Appendix A Out-of-distribution evaluation: both surrogates

Out-of-distribution performance was evaluated by constructing OOD sub-sets from the held-out test set: for each input feature in turn, samples with feature values in the upper or lower ∼20% of the training marginal were grouped as the OOD set; the remaining samples formed the in-distribution comparison. Table A1 reports test-set R^2 for coupled steady-state maximum global stress and k_{eff} under each OOD feature, for both Baseline and Regularized.

The largest Regularized improvement is for α_{base} extrapolation (stress $\Delta R^2 = +0.170$), consistent with the enforcement of the α_{base}–stress monotonicity constraint. For k_{eff} under α_{base} extrapolation, both models show substantial degradation ($R^2 \approx 0.29$), reflecting that extreme values of the dominant reactivity driver carry the model far outside the training regime; the monotonicity regularisation does not appreciably help in this case because the α_{base}–keff relationship is nonlinear over the extrapolation range. For ν, Regularized shows a slight stress R^2 decrease relative to Baseline (-0.005); the monotonicity constraint for this pair is less tight, and the difference is within expected sampling variability.

Table A1 OOD evaluation: coupled steady-state maximum global stress R^2 and k_{eff} R^2 for Baseline and Regularized surrogates, by OOD input feature. ΔR^2 = Regularized − Baseline (stress).

OOD feature	n_{OOD}	Stress R^2		k_{eff} R^2	
		Baseline	Regularized	Baseline	Regularized
$E_{\text{intercept}}$	577	0.940	0.949	0.814	0.826
α_{base}	580	0.774	**0.944**	0.286	0.285
ν	575	0.922	0.917	0.590	0.623
α_{slope}	576	0.909	0.934	0.708	0.706

Source: `paper_ood_multi_feature_summary.csv`. OOD sets cover the upper/lower ∼20% tail of the training marginal for each feature. Four features were selected to cover the four primary monotone constraints embedded in the Regularized objective.

Appendix B Baseline surrogate: full test-set comparison

Table B2 compares Baseline and Regularized on the held-out test set ($n = 435$) for the five primary outputs. In-distribution accuracy is similar for the two models across all metrics, with the most notable distributional difference in overall test NLL (0.359 for Baseline vs. 0.305 for Regularized, a 15% reduction) and in k_{eff} PICP_{90} (0.956 vs. 0.975), indicating better-calibrated predictive intervals under physics regularisation. Stress R^2 is identical (0.929) and stress RMSE differs by less than 0.1 MPa. The primary motivation for selecting Regularized as the main model is therefore its improved calibration (lower NLL) and substantially better OOD generalisation (Appendix A), not in-distribution point accuracy.

Table B2 Comparison of Baseline (baseline) and Regularized (physics-regularized) on the held-out test set ($n = 435$), primary outputs only. Test NLL: Baseline = 0.359, Regularized = 0.305.

Output		R^2		PICP_{90}	
	Pass	Baseline	Regularized	Baseline	Regularized
k_{eff}	Coupled	0.865	0.874	0.956	0.975
Max. fuel temp. (K)	Coupled	0.580	0.584	0.885	0.892
Max. monolith temp. (K)	Coupled	0.579	0.589	0.894	0.899
Max. global stress (MPa)	Coupled	0.929	0.929	0.931	0.913
Wall expansion (mm)	Coupled	0.996	0.996	1.000	0.989

Source: `paper_metrics_per_dim_level0.csv` and `paper_metrics_per_dim_level2.csv`. RMSE for max. global stress: Baseline = 7.93 MPa, Regularized = 7.91 MPa.

Appendix C Additional threshold analysis

Table C3 reports the posterior predictive failure probability at all three design thresholds for both Baseline and Regularized.

Table C3 Predicted failure probability $P_{\text{fail}}(\tau)$ at three stress thresholds (epistemic, 20 000 samples).

Level	$P(\sigma > 110\,\text{MPa})$	$P(\sigma > 120\,\text{MPa})$	$P(\sigma > 131\,\text{MPa})$
Baseline	1.000	1.000	1.000
Regularized	0.979	0.938	0.847

Source: `paper_safety_threshold_summary.csv`. Baseline values at 110 and 120 MPa exceed 1.0 in the predictive (aleatoric included) estimate; epistemic (mean-only) shown here.

Appendix D Full Sobol results

Table D4 provides first-order and total-order Sobol indices for all input–output pairs for both surrogate levels.

Table D4 First-order (S_1) Sobol indices for coupled steady-state maximum global stress and k_{eff}, Baseline and Regularized. Entries with CI including zero are marked with (†).

Output	Factor	S_1 (L0)	S_1 (L2)	L2 CI low	L2 CI high
Stress	E_{slope}	0.053	0.045	0.041	0.048
	$E_{\text{intercept}}$	0.613	0.598	0.596	0.601
	ν	0.051	0.051	0.047	0.054
	α_{base}	0.157	0.148	0.144	0.151
	α_{slope}	0.024	0.026	0.023	0.029
	$T_{\text{ref,SS316}}$	0.024	0.021	0.018	0.025
	$k_{\text{ref,SS316}}$	0.065	0.080	0.077	0.083
	α_{SS316}	$0.001^{\dagger}$	$0.002^{\dagger}$	−0.002	0.005
k_{eff}	E_{slope}	$0.003^{\dagger}$	$0.004^{\dagger}$	0.000	0.007
	$E_{\text{intercept}}$	0.004	$0.003^{\dagger}$	0.000	0.006
	ν	0.016	0.033	0.029	0.036
	α_{base}	0.809	0.775	0.774	0.776
	α_{slope}	0.162	0.177	0.174	0.180
	$T_{\text{ref,SS316}}$	0.006	$0.004^{\dagger}$	0.000	0.007
	$k_{\text{ref,SS316}}$	0.009	0.007	0.004	0.010
	α_{SS316}	0.006	0.005	0.002	0.008

Source: `experiments_0404/sensitivity/{baseline,data-mono}/sobol_results.csv`. L0 = baseline surrogate; L2 = data-mono (physics-regularized) surrogate.

Appendix E Per-output accuracy table (all 15 outputs)

The full per-output accuracy metrics for the Regularized surrogate on the test set are provided in Table E5.

Table E5 Complete per-output accuracy metrics, Regularized surrogate, test set ($n = 435$). Primary outputs marked ($\star$).

Output	Pass	Group	MAE	RMSE	R^2	PICP_{90}
Avg. fuel temp. (K)	Decoupled	sec	3.08	4.02	0.820	0.894
Max. fuel temp. (K)	Decoupled	sec	5.49	7.09	0.813	0.897
Max. monolith temp. (K)	Decoupled	sec	6.21	8.04	0.819	0.885
Max. global stress (MPa)	Decoupled	sec	8.84	11.25	0.928	0.920
Monolith new temp. (K)	Decoupled	sec	1.04	1.36	0.823	0.901
Core height after (mm)	Decoupled	sec	0.017	0.021	0.987	1.000
Wall expansion (mm)	Decoupled	sec	1.2×10^{-3}	1.5×10^{-3}	0.996	0.995
k_{eff} ($\star$)	Coupled	pri	2.0×10^{-4}	2.5×10^{-4}	0.874	0.975
Avg. fuel temp. (K)	Coupled	sec	1.89	2.43	0.584	0.887
Max. fuel temp. (K) ($\star$)	Coupled	pri	3.52	4.52	0.584	0.892
Max. monolith temp. (K) ($\star$)	Coupled	pri	3.96	5.07	0.589	0.899
Max. global stress (MPa) ($\star$)	Coupled	pri	6.10	7.91	0.929	0.913
Monolith new temp. (K)	Coupled	sec	0.51	0.67	0.598	0.897
Core height after (mm)	Coupled	sec	0.070	0.607	0.059	0.991
Wall expansion (mm) ($\star$)	Coupled	pri	1.3×10^{-3}	1.6×10^{-3}	0.996	0.989

Source: `paper_metrics_per_dim_level2.csv`. pri: primary output; sec: secondary output.

Appendix F Speedup benchmark details

The computational speedup of the Regularized surrogate versus the coupled HF solver is summarised in Table F6.

Table F6 Computational speedup of the surrogate versus the HF solver.

Metric	Value	Notes
Single-sample CPU latency	1.55×10^{-4} s	Regularized, CPU
Per-sample GPU batch latency	1.70×10^{-7} s	Batch size 1024
HF solver (one case)	$\sim$2000 s	OpenMC + FEniCS, 1 CPU node (user-confirmed; verify with run log)
Speedup (single CPU sample)	23.2×10^{6}	
Speedup (GPU batch per sample)	21.2×10^{9}	

Source: `paper_speedup_benchmark.json`.

Figure A1 | MCMC posterior traces (4 representative cases)

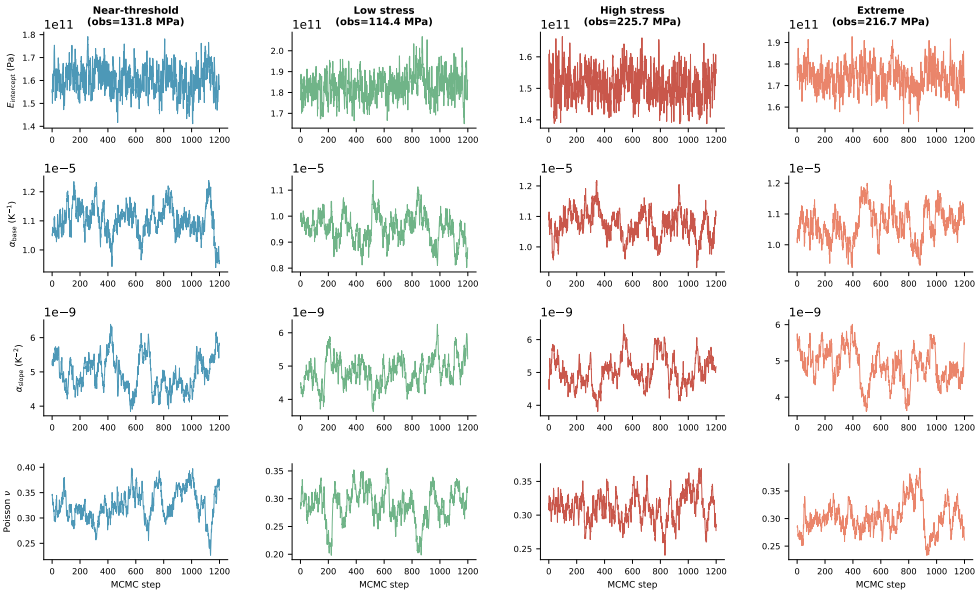

Fig. E1 MCMC posterior trace plots for four representative benchmark cases: near-threshold (observed ≈ 132 MPa), low stress (≈ 114 MPa), high stress (≈ 226 MPa), and extreme (≈ 217 MPa). Rows correspond to the four calibrated parameters ($E_{\text{intercept}}$, α_{base}, α_{slope}, ν); columns to cases. Visual inspection confirms adequate chain mixing in all cases (acceptance rates 0.47–0.61).

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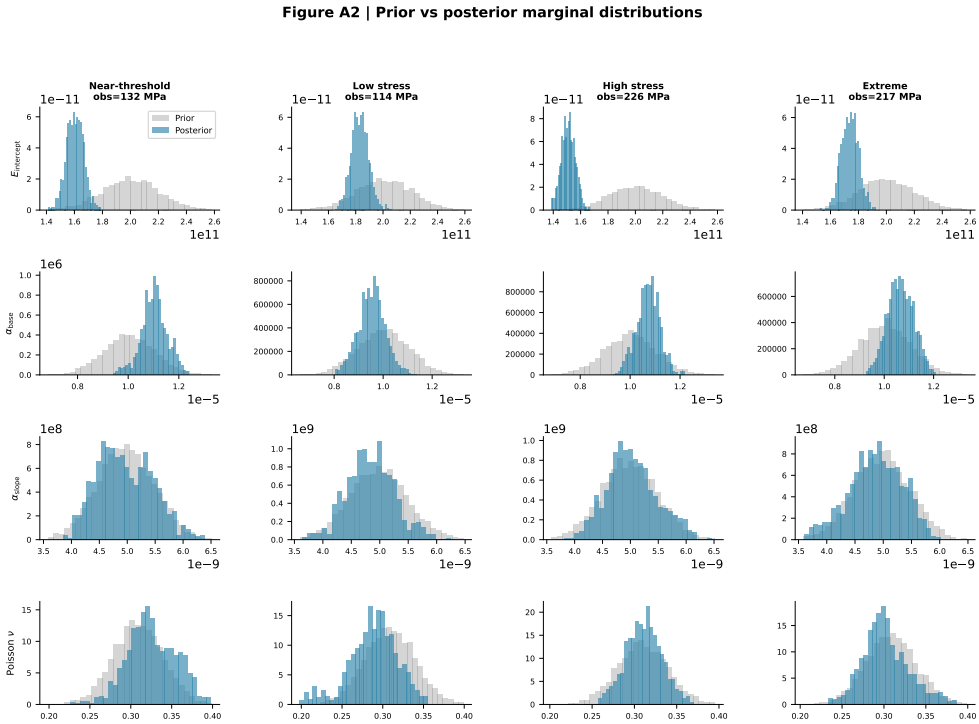

Fig. E2 Prior (grey) vs. posterior (blue) marginal distributions for the four calibrated parameters across four representative benchmark cases. Rows correspond to parameters; columns to cases (labelled with observed stress). Posterior contraction is most pronounced for α_{base} in the high-stress and extreme-stress cases, consistent with the Sobol analysis identifying α_{base} as the dominant reactivity driver.

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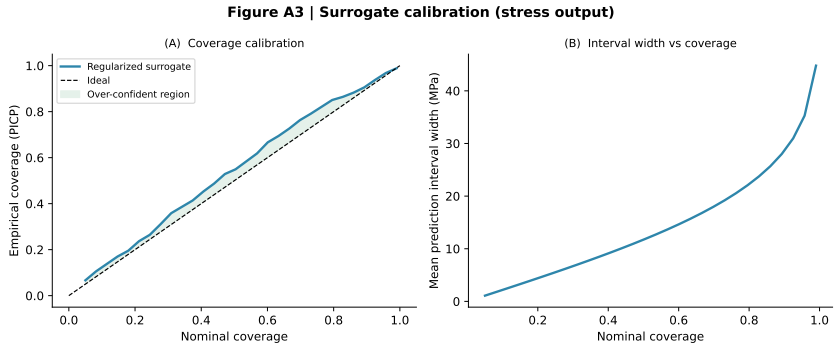

Fig. E3 Surrogate calibration diagnostics for the coupled steady-state maximum global stress output (Regularized, test set $n = 435$). (A) Reliability diagram: empirical prediction interval coverage probability (PICP) vs. nominal confidence level; the curve closely follows the ideal diagonal (dashed), confirming well-calibrated predictive uncertainty. (B) Mean prediction interval width as a function of nominal coverage.

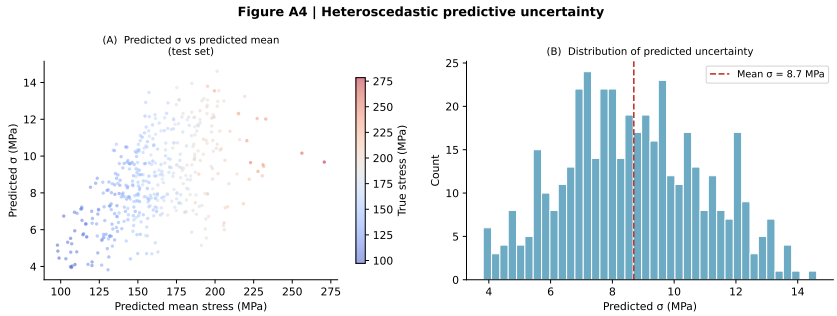

Fig. E4 Heteroscedastic predictive uncertainty for the coupled steady-state maximum global stress (Regularized, test set). (A) Predicted standard deviation σ as a function of predicted mean, coloured by true stress; σ increases in the high-stress regime, reflecting the model's recognition of higher aleatoric uncertainty in extreme parameter regions. (B) Distribution of predicted σ across test samples.

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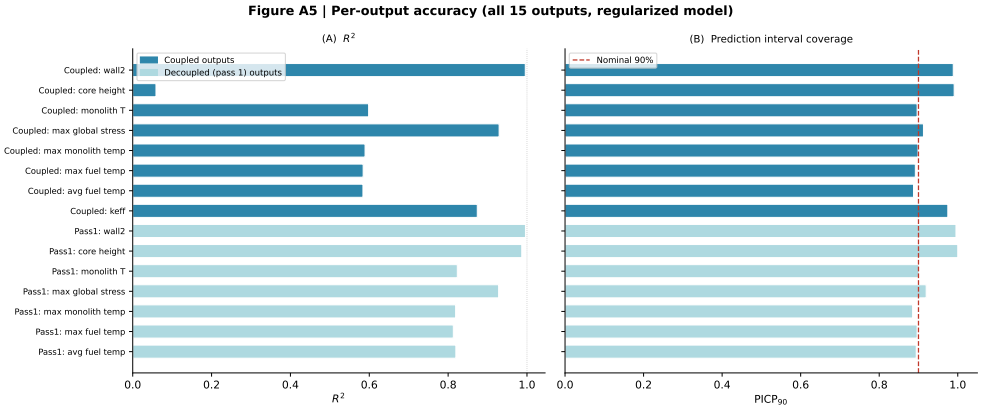

Fig. E5 Per-output accuracy for all 15 physics outputs (Regularized surrogate, test set). (A) R^2; (B) 90% prediction interval coverage (PICP_{90}). Blue bars: coupled steady-state outputs; light-blue bars: decoupled (decoupled single-pass) outputs. Wall expansion achieves $R^2 = 0.996$; coupled temperature outputs show lower $R^2 \approx 0.58$–0.60, reflecting their weaker sensitivity to the 8 material inputs in the coupled regime.

Monte Carlo. CRC Press, ??? (2011). https://doi.org/10.1201/b10905

Figure A6 | Prior vs posterior predictive distributions (4 cases)

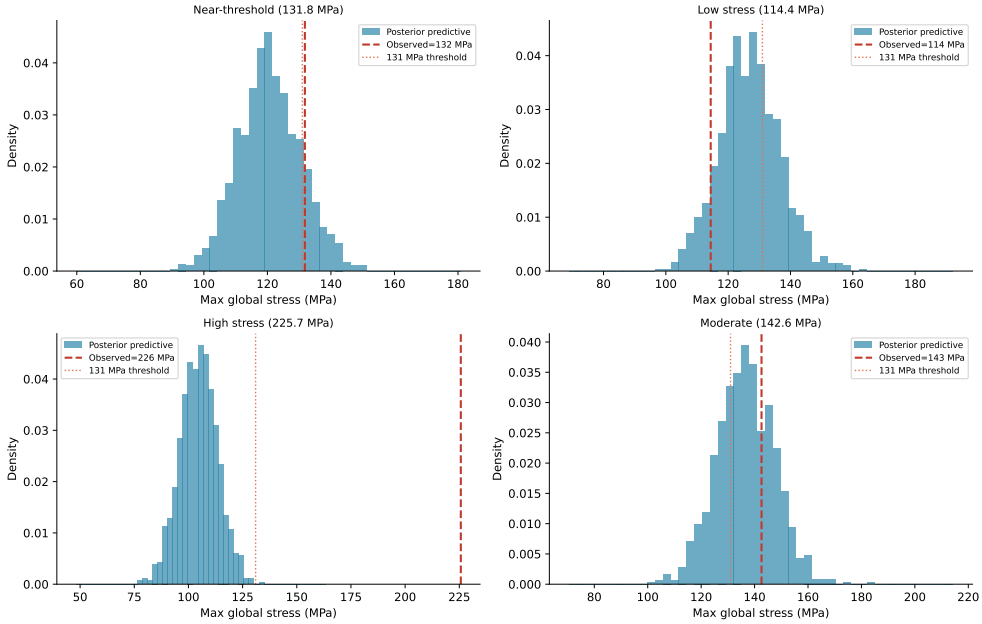

Fig. E6 Prior predictive (grey) vs. posterior predictive (blue) stress distributions for four representative benchmark cases (coupled steady-state max global stress). The observed stress (red dashed) and design threshold (orange dotted, 131 MPa) are indicated. Posterior updating concentrates probability mass near the observed value in all four cases, including the high-stress and extreme-stress scenarios.

Figure A7 | HeteroMLP architecture schematic

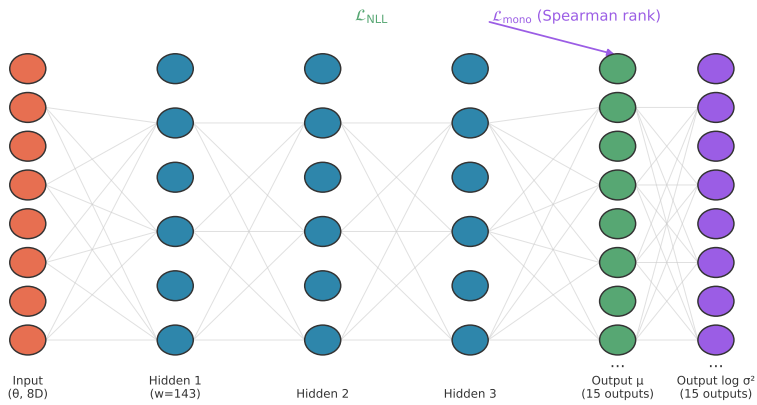

Fig. E7 Schematic of the heteroscedastic multilayer perceptron (HeteroMLP) architecture. The 8-dimensional material parameter vector $\boldsymbol{\theta}$ passes through shared hidden layers (width 143, depth 3 for Regularized) and is decoded into two heads: a mean output $\boldsymbol{\mu}$ and a log-variance output $\log \boldsymbol{\sigma}^2$, each of dimension 15. The Spearman-rank monotonicity regularization $\mathcal{L}_{\text{mono}}$ (Eq. 2) is applied during training via the log-variance head.

Figure A8 | Training curves (NLL vs epoch)

□□□□□ Re-run run_train_fixed_surrogates.py to generate training_history_level{0,2}.csv (now supported by the updated script).

Fig. E8 Training curves: negative log-likelihood (NLL) vs. epoch for the baseline (Baseline) and physics-regularized (Regularized) surrogates. (*Placeholder shown; regenerate with actual training history files once available by re-running* `run_train_fixed_surrogates.py`.)